

Reading the results

A guide to what the pipeline does, how to interpret what comes out of it, and what it cannot tell you.

One correction before anything else. A p-value is never negative. It is a probability, so it lives strictly between 0 and 1. What can be positive or negative is the **correlation, r**, which runs from -1 to +1. The two get conflated constantly and the difference matters: r is a *direction and a size*; p is a statement about *how often chance would do this well*. Section 3 works through this properly.

What is in here

Part	Question it answers
1 The pipeline	What actually happens to a C3D file, and which analysis answers which question
2 The metrics	What r, p, r-squared, ICC and LOO R-squared each mean <i>in this study</i>
3 The confusions	Negative p-values, high-r-but-not-significant, and why n = 30 changes the rules
4 Worked examples	Six real results from your data, read line by line
5 Limitations	What this dataset cannot establish, and why
6 Checklist	Red flags to run any new result through

Part 1 · What the pipeline does

Every trial goes through the same chain. Each step exists to remove something that would otherwise be mistaken for technique.

Stage	What it does	What it removes
Load	Read the C3D, gap-fill, convert mm to m	Missing frames, unit ambiguity
Align	SVD across all 64 markers to find the travel axis. Gravity-aligned Z is left alone	Lane drift and camera orientation
Cut lead-in	Detect the block-set frame (both hands down)	Standing-around before the start
Origin + trim	Rear heel to (0,0,0); keep the first 62 m	Track position
Segment	Contact-to-contact stride windows, validated 22-40 frames	Merged or partial strides
Normalise	Resample each stride to 101 points, 0-100 % of the cycle	Stride-frequency differences
Angles	13 sagittal joint angles from segment-to-segment turns	Body size (an angle has no units of length)

Four analyses, four different questions

These are **separate**. A result from one is not a result from another, and the most common way to misread this project is to blend them.

Analysis	Asks	Output
fPCA	How do these athletes differ from one another?	Components ranked by <i>variance</i> . Unsupervised: it never sees velocity.
Angle correlations	Does one summary number track speed?	One r per feature. This is where 'knee ROM' comes from - not from the PCA.
Gait timing	Do contact and flight time track speed?	Contact time is the strongest single correlate in the project.
LOO models	Can we predict a held-out athlete's speed?	Cross-validated R-squared. The only number that speaks to prediction.

The single most important structural fact. The fPCA is *unsupervised*: it finds directions of greatest variance, which need not be the directions that predict speed. In this cohort they are not. Across all six components (81 % of variance) the largest correlation with knee ROM is 0.213 and with contact time 0.372. **The PCA does not find what predicts speed, and it was never going to.** That is a property of the method, not a bug in the data.

Part 2 · What each metric means here

r — the correlation

Runs from -1 to +1. It has two parts and they are independent of each other:

Part	Reading
Sign	Negative: as the feature goes up, velocity goes down. <i>Larger knee ROM, slower athlete.</i> Positive: they rise together. <i>Higher stride frequency, faster athlete.</i>
Magnitude	How tightly the points hug a straight line. 0 is a shapeless cloud, 1 is a perfect line. It says nothing about the <i>slope</i> - a steep relationship and a shallow one can have the same r.

A negative r is not a worse result than a positive one. It is a different direction, nothing more. 'Less of this goes with more speed' is as real a finding as the reverse.

r-squared — the share of variance explained

Square r and you get the proportion of between-athlete variance in top speed that the feature accounts for. This is the honest size of an effect, and it is always smaller than r looks: **r = -0.552 becomes r-squared = 0.305**, so knee ROM accounts for about 30 % of the differences in speed and leaves 70 % elsewhere.

p — how often chance would do this well

The probability of seeing a correlation at least this strong if there were truly no relationship, given this many athletes. **p = 0.0016 means about 1 run in 600.** Three things it is *not*:

p is not...	Because
the size of the effect	That is r-squared. A tiny effect can have a tiny p if n is large.
the probability the finding is wrong	It assumes there is no effect and asks how surprising the data would be.
ever negative	It is a probability. 0 to 1, always.

ICC(1,1) — how much of the measurement is the athlete

Each athlete contributes five separate strides. ICC asks how much of a feature's variation is *between athletes* rather than *between repeat strides of the same athlete*. It is a reliability score, and it never looks at velocity - which is why screening features on it cannot leak.

ICC	Example here	What to do with it
0.89	trunk lean mean	Stable trait. Safe to model.
0.85	peak hip extension	Good. Correlations mildly attenuated.
0.57	ankle_lo_mean	Half is stride noise. Correlations pulled toward zero.
0.48	knee_hi_mean	Mostly noise. Do not put it in a between-athlete model.

Low ICC does not mean the joint is unimportant. It means *this measurement of it, at 60 Hz, on five strides* is too noisy to carry between-athlete signal.

LOO R-squared — can it predict someone new

Fit on 29 athletes, predict the 30th, repeat 30 times, and score only those held-out predictions. Unlike r, it can go **negative**, and that has a plain reading: the model is worse than simply guessing the cohort mean. The honest zero line here is **-0.070**, not 0.000, because leave-one-out makes even the mean slightly worse than the full-sample mean.

Part 3 · The cases you asked about

At $n = 30$, r and p carry the same information

This is the key point, and it dissolves most of the confusion. With the sample size fixed, p is a **deterministic function of r** . There is one threshold:

At $n = 30$, $p < 0.05$ happens exactly when $|r| > 0.361$ (two-tailed), or $|r| > 0.306$ for a one-tailed test where the direction was predicted in advance.

So within this study:

Case	Can it happen at $n = 30$?
High r , not significant	No - not above $ r = 0.361$. If $r = 0.55$ it is significant, necessarily.
Low r , significant	No - not below $ r = 0.361$. If $r = 0.20$ it is not significant, necessarily.

Those two cases *only* arise when comparing across **different sample sizes**. That is where the intuition comes from, and it is worth seeing why:

n	$ r $ needed for $p < 0.05$	What that means
10	0.632	Only huge effects clear the bar. Real ones get missed.
30	0.361	Your study.
100	0.197	Modest effects clear it.
500	0.088	Trivial effects clear it. Significance stops being informative.

This is why a large study can report a 'highly significant' $r = 0.09$ that means almost nothing (r -squared = 0.008), and why your $n = 30$ can miss a real effect of $r = 0.30$ entirely. **Report r , r -squared and the confidence interval. p alone is the least informative of the four.**

The confidence interval is what $n = 30$ really costs

At this sample size the intervals are wide enough that the magnitude is barely pinned down even when the direction is solid:

r	p (two-tailed)	r -squared	95 % CI on r
0.20	0.289	0.04	-0.17 to +0.52
0.361	0.050	0.13	0.00 to +0.64 (<i>the threshold</i>)
0.50	0.005	0.25	+0.17 to +0.73
0.552	0.002	0.31	+0.24 to +0.76 (<i>knee ROM</i>)
0.638	0.0001	0.41	+0.36 to +0.81 (<i>contact time</i>)

Read the knee ROM row honestly: the direction is secure, but the true correlation could plausibly be anywhere from 0.24 to 0.76 - which is the difference between a minor factor and a dominant one.

Searching many features breaks the threshold

The 0.361 threshold is only valid for a test chosen **before** looking. Search 23 features and take the biggest, and the rules change completely. Shuffling velocity 4000 times and recording the largest of 23 correlations each time:

Largest $ r $ of 23, under the null	Value
median	0.378

Largest r of 23, under the null	Value
95th percentile	0.525
observed in your data	0.597 → corrected p = 0.009

The median null maximum is **0.378**, which is already above the nominal 0.361 threshold. **Searching 23 features, the biggest correlation clears 'significance' by luck more than half the time.** Knee ROM survives this properly corrected test - but only because it was checked against the right null, not the textbook one.

Part 4 · Six real results, read line by line

Result	How to read it
Contact time r = -0.638, p = 0.00015, r-squared = 0.41	Strongest single correlate in the project. Negative: shorter contact, faster athlete . Explains ~41 % of between-athlete variance. Reproduces Weyand et al. (2000).
Flight time r = -0.041, p = 0.83	A clean null. Flight time is essentially constant across athletes, so the stride-frequency gain comes <i>entirely</i> from shortening contact. A null result is a finding , not a failure.
Knee ROM r = -0.552, p = 0.0016, r-squared = 0.31	Negative: larger knee range of motion, slower athlete . Survives controls for stride duration and contact time. But it was <i>found by searching</i> , so it needs the corrected null above.
fPCA PC1 42.0 % of variance, r = -0.177	Explains the most variance and predicts almost nothing. It is trunk-lean posture. Variance explained and predictive value are unrelated quantities.
Best model LOO R-squared = +0.271, permutation p = 0.002	Four features predicting held-out athletes, against a -0.070 null. Genuine but modest. RMSE 0.61 m/s against a cohort spread of 0.72 m/s.
All 23 features, ridge LOO R-squared = -0.244	Worse than guessing the mean. With 23 features and 30 athletes the model fits noise. This is why the four-feature model is the honest one.

Part 5 · Limitations

Limitation	Consequence
n = 30 is the real n	Confirmed twice. Velocity ICC across an athlete's own strides is 0.995, so the 150 strides are 30 clusters of near-identical replicates. Resampling strides cannot buy statistical power.
60 Hz is too slow for contact	Ground contact lasts 5.6 frames. One frame is 18 % of the phase. Any feature divided by contact time inherits that: ICC falls from 0.51 to 0.33. 240 Hz+ would fix it.
No force plate	Contact is inferred from marker height, not measured. The threshold carries roughly +/-25 % uncertainty. The published standard method (Zeni 2008) fails here entirely because it assumes heel-strike walking and sprinters are forefoot strikers.
Phase-windowed features are confounded	Any range of motion measured over a window whose <i>length</i> depends on speed will correlate with speed mechanically. Seven stance-phase findings collapsed on this. Always control for window length.
Absolute angles are not comparable to the literature	The hip is measured against the trunk rather than the pelvis; peak knee flexion reads ~17 degrees below published video work. Relative differences between athletes are sound; absolute values are not.
Everything is correlational	No intervention, no randomisation. 'Faster athletes have shorter contact times' does not establish that shortening contact would make someone faster.

Part 6 · Checklist for any new result

Ask	Why
Was this hypothesis chosen before or after looking?	Decides whether $p = 0.05$ or the corrected null applies.
What is r-squared, and what is the 95 % CI?	r and p alone overstate how much is known.
What is the feature's ICC?	Below 0.5 the measurement is mostly stride noise.
Does the window length depend on the outcome?	If yes, the correlation may be arithmetic, not biomechanics.
Does it survive controlling for height and contact time?	Those two explain a great deal on their own.
Does one athlete carry it?	At $n = 30$, check Cook's distance against the $4/n = 0.133$ flag.
Does it hold out of sample?	In-sample fit is not prediction. Use the LOO number.

Every figure in this document is reproduced by a labelled cell in `04_Audit_Provenance.ipynb`. The full audit, including the collinearity diagnostics and the component-by-component interpretation, is in the Sprint Pipeline Audit.